

Math 252, 10.3, Calculus of Parametric Equations

1. **Motivation.** Consider again, a ball is thrown so that its path $x = 24\sqrt{2}t$ and $y = -16t^2 + 24\sqrt{2}t$, where t is time in seconds. How could we determine its velocity and acceleration?

The horizontal and vertical components of velocity are

$$\frac{dx}{dt} = 24\sqrt{2} \quad \text{and} \quad \frac{dy}{dt} = -32t + 24\sqrt{2}.$$

Thus, the velocity vector is

$$\mathbf{v}(t) = \langle 24\sqrt{2}, -32t + 24\sqrt{2} \rangle.$$

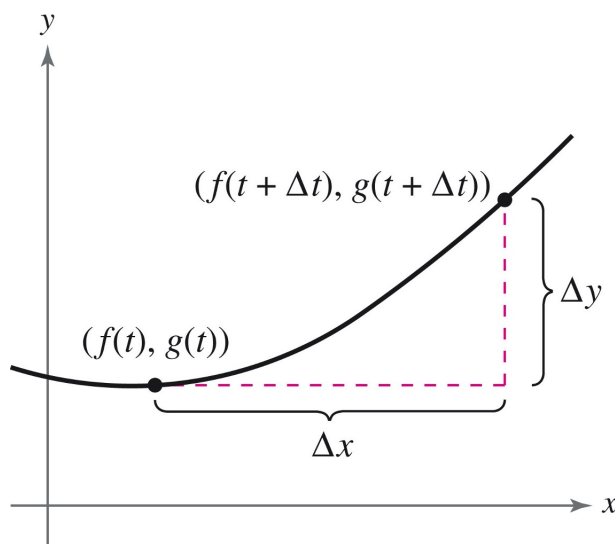
The horizontal and vertical components of acceleration are

$$\frac{d^2x}{dt^2} = 0 \quad \text{and} \quad \frac{d^2y}{dt^2} = -32.$$

Thus, the acceleration vector is

$$\mathbf{a}(t) = \langle 0, -32 \rangle.$$

2. Derivation of Derivative Formulas



Suppose that a parametric curve is given by

$$x = f(t) \quad \text{and} \quad y = g(t).$$

Consider the points on the curve corresponding to the parameter values t and $t + \Delta t$:

$$P = (f(t), g(t)) \quad \text{and} \quad Q = (f(t + \Delta t), g(t + \Delta t)).$$

The slope of the secant line through P and Q is

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{g(t + \Delta t) - g(t)}{f(t + \Delta t) - f(t)} \\ &= \frac{\frac{g(t + \Delta t) - g(t)}{\Delta t}}{\frac{f(t + \Delta t) - f(t)}{\Delta t}}. \end{aligned}$$

As $\Delta t \rightarrow 0$, the point Q approaches the point P , and the secant line approaches the tangent line. Therefore,

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta t \rightarrow 0} \frac{\frac{g(t + \Delta t) - g(t)}{\Delta t}}{\frac{f(t + \Delta t) - f(t)}{\Delta t}} \\ &= \frac{\lim_{\Delta t \rightarrow 0} \frac{g(t + \Delta t) - g(t)}{\Delta t}}{\lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}} \\ &= \frac{g'(t)}{f'(t)} \\ &= \frac{dy/dt}{dx/dt},\end{aligned}$$

provided $\frac{dx}{dt} \neq 0$.

3. **Example 1:** Let $x = \sin t$ and $y = \cos t$. Find dy/dx

$$\frac{dx}{dt} = \cos t \quad \text{and} \quad \frac{dy}{dt} = -\sin t.$$

Therefore,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-\sin t}{\cos t} = -\tan t.$$

4. Higher Order Derivatives.

We know that

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

To find the second derivative, differentiate the first derivative with respect to x :

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right).$$

Using the first derivative formula with

$$y = \frac{dy}{dx},$$

we obtain

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}.$$

Since

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt},$$

it follows that

$$\boxed{\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy/dt}{dx/dt} \right)}{\frac{dx}{dt}}}.$$

5. **Example 2:** For the curve $x = \sqrt{t}$ and $y = \frac{1}{4}(t^2 - 4)$, for $t \geq 0$, find the slope and concavity at the point $(2, 3)$.

At the point $(2, 3)$,

$$2 = \sqrt{t},$$

so

$$t = 4.$$

Now,

$$\frac{dx}{dt} = \frac{1}{2\sqrt{t}} \quad \text{and} \quad \frac{dy}{dt} = \frac{t}{2}.$$

Thus,

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} \\ &= \frac{t/2}{1/(2\sqrt{t})} \\ &= t\sqrt{t}. \end{aligned}$$

At $t = 4$,

$$\frac{dy}{dx} = 4\sqrt{4} = 8.$$

Therefore, the slope at $(2, 3)$ is 8.

For concavity,

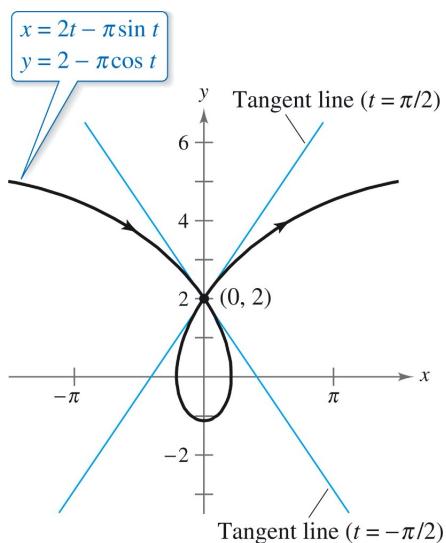
$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} \\ &= \frac{\frac{d}{dt} (t^{3/2})}{1/(2\sqrt{t})} \\ &= \frac{\frac{3}{2}\sqrt{t}}{1/(2\sqrt{t})} \\ &= 3t. \end{aligned}$$

At $t = 4$,

$$\frac{d^2y}{dx^2} = 12 > 0.$$

Therefore, the curve is concave upward at $(2, 3)$.

6. **Example 3:** The prolate cycloid is given by, $x = 2t - \pi \sin t$ and $y = 2 - \pi \cos t$. The curve crosses itself at the point $(0, 2)$, as shown. Find the equation of the tangent lines at this point.



At the point $(0, 2)$,

$$2 - \pi \cos t = 2,$$

so

$$\cos t = 0.$$

The two parameter values at which the curve passes through $(0, 2)$ are

$$t = \frac{\pi}{2} \quad \text{and} \quad t = -\frac{\pi}{2}.$$

Indeed,

$$2 \left(\frac{\pi}{2} \right) - \pi \sin \left(\frac{\pi}{2} \right) = 0$$

and

$$2 \left(-\frac{\pi}{2} \right) - \pi \sin \left(-\frac{\pi}{2} \right) = 0.$$

Now,

$$\frac{dx}{dt} = 2 - \pi \cos t \quad \text{and} \quad \frac{dy}{dt} = \pi \sin t.$$

Thus,

$$\frac{dy}{dx} = \frac{\pi \sin t}{2 - \pi \cos t}.$$

At $t = \frac{\pi}{2}$,

$$\frac{dy}{dx} = \frac{\pi}{2}.$$

Therefore, one tangent line is

$$y - 2 = \frac{\pi}{2}x.$$

At $t = -\frac{\pi}{2}$,

$$\frac{dy}{dx} = -\frac{\pi}{2}.$$

Therefore, the other tangent line is

$$y - 2 = -\frac{\pi}{2}x.$$

7. **Arc Length** Consider a curve C whose equation is given by $y = h(x)$. If we want to find the length of the arc between x_0 and x_1 , recall we have the formula

$$\begin{aligned} s &= \int_{x_0}^{x_1} \sqrt{1 + [h'(x)]^2} dx \\ &= \int_{x_0}^{x_1} \sqrt{1 + [dy/dx]^2} dx \end{aligned}$$

Suppose that $x = f(t)$ and $y = g(t)$, where $t_0 \leq t \leq t_1$. Since

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \text{and} \quad dx = \frac{dx}{dt} dt,$$

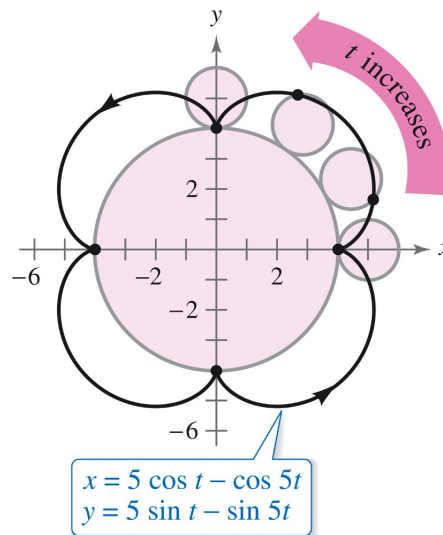
we have

$$\begin{aligned} s &= \int_{t_0}^{t_1} \sqrt{1 + \left[\frac{dy/dt}{dx/dt} \right]^2} \frac{dx}{dt} dt \\ &= \int_{t_0}^{t_1} \sqrt{\frac{(dx/dt)^2 + (dy/dt)^2}{(dx/dt)^2}} \frac{dx}{dt} dt \\ &= \int_{t_0}^{t_1} \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt. \end{aligned}$$

Therefore, the arc length of a parametric curve is

$$s = \int_{t_0}^{t_1} \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt.$$

8. **Example 4:** A circle of radius 1 rolls around the circumference of a large circle of radius 4. If a point on the smaller circle is traced, the result is called an epicycloid, and its equation is given parametrically by $x = 5 \cos t - \cos 5t$ and $y = 5 \sin t - \sin 5t$. Determine the distance traveled by the point in one complete trip around the larger circle.



One complete trip around the larger circle occurs for

$$0 \leq t \leq 2\pi.$$

The epicycloid has four-fold rotational symmetry. The portion of the curve traced for

$$0 \leq t \leq \frac{\pi}{2}$$

has the same length as each of the other three portions of the curve. Therefore, we find the length of this portion and multiply by 4.

We have

$$\frac{dx}{dt} = -5 \sin t + 5 \sin 5t$$

and

$$\frac{dy}{dt} = 5 \cos t - 5 \cos 5t.$$

Therefore,

$$\begin{aligned} s &= 4 \int_0^{\pi/2} \sqrt{(-5 \sin t + 5 \sin 5t)^2 + (5 \cos t - 5 \cos 5t)^2} dt \\ &= 20 \int_0^{\pi/2} \sqrt{(\sin 5t - \sin t)^2 + (\cos t - \cos 5t)^2} dt \\ &= 20 \int_0^{\pi/2} \sqrt{2 - 2 \cos 4t} dt \\ &= 20 \int_0^{\pi/2} \sqrt{4 \sin^2 2t} dt. \end{aligned}$$

Since

$$0 \leq t \leq \frac{\pi}{2},$$

we have

$$0 \leq 2t \leq \pi,$$

and hence $\sin 2t \geq 0$. Therefore,

$$\sqrt{4 \sin^2 2t} = 2 \sin 2t.$$

Thus,

$$\begin{aligned} s &= 40 \int_0^{\pi/2} \sin 2t dt \\ &= 40 \left[-\frac{1}{2} \cos 2t \right]_0^{\pi/2} \\ &= 40. \end{aligned}$$

Therefore, the point travels a distance of

$$\boxed{40}.$$